Pratap Chandra Deo

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Professional Summary

Full-Stack Engineer with 3 years of expertise delivering scalable enterprise solutions across education, government, and corporate sectors. Specialized in AI integration, system modernization, and performance optimization, with demonstrated success in significantly reducing operational costs and improving system efficiency.. Proven track record in leading technical initiatives from concept to deployment, with deep expertise in cloud architecture, database design, and full-stack development.

Technical Skills

Core Languages: Java, Python, JavaScript, HTML5, CSS3
Frontend: React, Bootstrap, jQuery, Redux, Material-UI
Backend & APIs: Node.js, Express, Spring Boot, Django, REST APIs, GraphQL, Microservices
Databases: PostgreSQL, MySQL, MongoDB, Redis, Database Design & Optimization
Cloud & DevOps: AWS (EC2, S3, Lambda), Azure, Docker, Kubernetes, GitHub Actions, CI/CD, Firebase
AI & Advanced: LLM Integration, Speech-to-Text, WebSockets, Real-time Communication
Tools & Methods: Git, JIRA, Postman, Swagger, Agile/Scrum, System Design, TDD

Experience

Software Engineer | *NIT Warangal* *Aug 2024 – Present*

- Leading digital transformation initiatives across three major systems (ERP, Recruitment, Website) serving a large user base, improving administrative processes by around 30%
- Architected microservices-based ERP enhancements handling hundreds of daily workflows, reducing manual processing time by roughly one-third through automation
- Modernized legacy systems by upgrading Node.js infrastructure, resulting in notable performance improvement and near-continuous system availability
- Mentored junior developers and implemented CI/CD practices, lowering deployment issues and improving team delivery speed

Full Stack Developer | *Mechanetics Solutions Pvt. Ltd.* *Feb 2022 – Jul 2024*

- Engineered several scalable applications using React and Spring Boot, supporting thousands of active users with high uptime and fast response times
- Spearheaded BMC Garbage Management System that reduced complaint resolution time from two days to a few hours, achieving strong user feedback
- Architected JSW Load Management System optimizing significant operational costs through predictive resource allocation and live monitoring
- Established DevOps pipelines that reduced deployment cycles substantially and improved code test coverage to a stable level
- Introduced Agile methods improving team velocity and shortening delivery timelines across multiple client projects

Open Source Contributions

Kestra | *Event-Driven Declarative Orchestration Platform* *PR #12108*

- Fixed critical UI rendering issue by converting `::alert{type="..."}` syntax to standard markdown containers in plugin documentation
- Enhanced documentation readability by implementing regex conversion for alert blocks, resolving raw markdown display issues affecting user experience
- Registered all alert container types (warning, info, danger, success, tip) with markdown-it parser to ensure proper styling across the platform
- Handled both multi-line and single-line alert syntax while maintaining proper formatting and whitespace preservation
- Contribution was thoroughly reviewed, approved, and merged into the main codebase, fixing GitHub issue #12019

PROJECTS

Safe Line - AI-Powered Cybercrime Reporting System

Link

- Pioneered AI-driven voice agent reducing emergency response time from 5 minutes to around 15 seconds, serving over 1,000 simulated incidents with about 90% classification accuracy
- Engineered real-time speech processing system achieving nearly 90% transcription accuracy and around 98% SMS delivery reliability, automating roughly 80% of manual data entry
- Architected scalable microservices capable of handling hundreds of concurrent sessions, implementing detailed audit trails and emergency prioritization logic **Projected Impact:** Expected to lower operational costs by nearly 40% and enhance citizen satisfaction based on pilot simulations

Tech Stack: Python, Flask, LiveKit, PostgreSQL, Deepgram API, Cerebras API, Vonage API

Smile - ERP System Application

Link

- Enhanced internal ERP system managing hundreds of administrative workflows across departments, serving thousands of users
- Developed multiple new modules expanding system functionality, reducing manual process time by around 30% and improving data accuracy significantly
- Optimized existing modules based on user feedback, improving overall system performance by roughly 20% and user satisfaction levels
- Collaborated with stakeholders to deliver tailored solutions, saving several hundred administrative hours monthly

Tech Stack: Java, jQuery, PostgreSQL, AJAX, Hibernate

NITW Website Development

Link

- Led end-to-end development of NITW's official website serving thousands of monthly visitors with strong uptime and reliability
- Engineered an admin panel managing hundreds of content pages, cutting content update time from hours to minutes
- Upgraded entire stack from Node.js 16 to 22.12 LTS, boosting performance by about 35% and resolving major compatibility issues
- Implemented a secure authentication system protecting user accounts with no reported breaches
- Optimized database queries and APIs, reducing average page load times notably

Tech Stack: JavaScript, Node.js (v22.12 LTS), React.js, Express.js, MongoDB, Git

Recruitment Portal - NIT Warangal

Link

- Built a comprehensive Recruitment Portal handling thousands of faculty applications annually, cutting hiring cycle time by about 50%
- Developed a dynamic form system supporting numerous data fields with accurate validation
- Implemented role-based access for admin users, improving review efficiency by nearly 40%
- Created an automated notification system ensuring reliable email delivery at scale
- Designed an optimized PostgreSQL database for fast responses on complex search queries

Tech Stack: Python3, Django, PostgreSQL, HTML, Crispy Forms

EDUCATION

BIJU PATNAIK UNIVERSITY	**Master of Computer Application** — CGPA: 8.1	*November 2020 – September 2022*
UTKAL UNIVERSITY	**Bachelor's in Computer Application** — CGPA: 7.2	*August 2017 – November 2020*